

JACOB HESS

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EXPERIENCE

Mothership — Lead QA and Website Developer

April 2025 – Present

- Led quality assurance testing processes and protocols to ensure product reliability and performance
- Developed and maintained company website using modern web development technologies
- Collaborated with cross-functional teams to identify and resolve software issues
- Implemented automated testing frameworks to improve efficiency and accuracy

Costco, Cedar Park — Major and Electronic Sales

June 2022 – January 2025

- Assisted customers in selecting and purchasing electronic products, specializing in computers, televisions, and other high-demand items
- Utilized in-depth knowledge of computer hardware and software to guide customers in making informed decisions

EDUCATION

Western Governors University, Austin — Bachelor of Cyber Security

January 2023 – May 2025 | Graduated May 2025

PROJECTS

Talk Radius — Local Voice Chat App

talkradius.com | 2026 – Present

- Built a proximity-based, live voice chat application that lets people in the same physical space talk in real time without accounts, profiles, or stored conversations
- Architected end-to-end encrypted, peer-to-peer audio over WebRTC with DTLS-SRTP; signaling server holds no key material and never sees audio
- Designed privacy model around ephemeral, server-issued peer IDs and coarse (~150m geohash) location, keeping precise GPS on-device
- Hardened signaling server with origin allowlisting, per-IP connection caps, per-connection rate limits, and bounded message sizes

sigma-pipeline — Detection-as-Code CI/CD for Sigma Rules

github.com/JacobRHess/sigma-pipeline | 2026

- Built Python CLI (`sigma lint | test | diff | deploy | coverage`) bringing fixture-driven testing to Sigma detection rules — proving rules fire on real threats and stay quiet on benign activity
- Designed deploy stage that registers rules as Splunk saved searches via the REST API; idempotent create-or-update so repeated deploys converge
- Wired GitHub Actions workflow with dry-run vs. apply modes and a manual-approval production gate; shipped a Splunk dashboard for MITRE ATT&CK coverage

splunk-sigma — Native Sigma Rule Evaluator for Splunk

github.com/JacobRHess/splunk-sigma | 2026

- Built an open-source Splunk app that adds a `| sigma` custom search command, evaluating Sigma YAML detection rules natively at search time instead of pre-converting them to static SPL
- Implemented a ~600-line Python interpreter: YAML loader, condition-language parser to AST, evaluator, and field-modifier operators
- Designed a dual front-end: in-Splunk `StreamingCommand` and a standalone service that consumes Splunk's REST API, both backed by the same engine; MIT-licensed with pytest CI

Bat Conservation Trust — Website Development

September 2024

- Resolved website performance issues by minimizing external HTTP requests and optimizing load times
- Updated security protocols to meet current regulations and industry best practices
- Implemented secure payment processing and reinforced SSL/TLS encryption for donation system

OWASP Amass — Documentation

June 2024

- Contributed detailed user instructions and comprehensive FAQs to OWASP Amass documentation
- Applied technical writing skills to explain cybersecurity concepts related to network discovery and reconnaissance

TECHNICAL SKILLS

Programming Languages: Python, Go, Rust, Swift, Java, C++, JavaScript

Detection Engineering: Sigma, Splunk SPL, MITRE ATT&CK, YARA-style rule authoring, fixture-driven rule testing

Security Tooling: Splunk (custom search commands, REST API, saved searches, dashboards), pySigma, OWASP Amass

DevSecOps: GitHub Actions, CI/CD, pytest, detection-as-code, code review workflows

Web Development: HTML, CSS, JavaScript, Responsive Design

Operating Systems & Tools: Linux, Bash, GitHub, Unity, Microsoft Office 365

Quality Assurance: Test Planning, Automated Testing, Bug Tracking

CERTIFICATIONS

CompTIA Security+ • CompTIA Network+ • CISSP – Associate • CCSP – Associate